



Modeling motorcycle crash frequency on rural multilane segments in Kentucky[☆]

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ABSTRACT

Despite motorcycle crashes accounting for a large percentage of traffic fatalities in the U.S., studies investigating motorcycle crash frequency are relatively limited. This study took the initiative and developed motorcycle crash-specific safety performance functions (SPFs) along rural multilane road segments in Kentucky, separately for the pre-COVID-19 pandemic (2015–2019) and post-COVID-19 pandemic (2020–2022) periods. Eight years of motorcycle crash records (2015 through 2022) and site-specific characteristics (e.g., shoulder width and annual average daily traffic “AADT”) were collected and used. Conway-Maxwell-Poisson (CMP) and heterogeneous Conway-Maxwell-Poisson (HTCMP) models were fitted and compared while accounting for motorcycle crash under-dispersion (i.e., when crash variance is less than its mean). The study results showed that, for both pre- and post-pandemic periods, the HTCMP models outperformed their CMP counterparts based on various goodness-of-fit measures (e.g., likelihood ratio test “LRT”, Akaike information criterion “AIC”, and McFadden pseudo R-squared) and prediction performance measures (i.e., mean absolute deviance “MAD” and mean square prediction error “MSPE”). From the developed SPFs, for the pre-pandemic period, the presence of horizontal curves and undivided roadways were significantly associated with increased motorcycle crash frequency along rural multilane segments, while in the post-pandemic period, wider right shoulders and higher AADT were significantly associated with increased motorcycle crash frequency. The predicted crash frequencies while applying the best-fit models (i.e., the HTCMP models) were then used to identify and rank high-crash rural multilane segments in Kentucky. Based on the study findings, several countermeasures were proposed to improve motorcyclists’ safety along Kentucky’s rural multilane segments, e.g., adding centerline grooved rumble strips along undivided rural multilane roadways.

1. Introduction

Motorcyclists are vulnerable road users with a heightened risk of traffic casualties due to the inherent instability of the motorcycles and the absence of in-vehicle protection. According to the National Highway Traffic Safety Administration (NHTSA), motorcycles contribute to 14 % of all traffic fatalities and 17 % of all motor vehicle occupant fatalities in the U.S., despite comprising only 3.5 % of registered vehicles (NHTSA, 2023). Furthermore, motorcycles accounted for over 7 % of traffic fatalities in Kentucky, while accounting for less than 1 % of registered vehicles (KTC, 2023). Moreover, the emergence of the 2019 novel coronavirus (COVID-19), resulted in significant changes in travel

behavior, travel patterns, and traffic safety (Chand et al., 2021; Yasin et al., 2021). For example, Islam et al. (2022) examined risk compensation trends in road safety during COVID-19 along Interstate 4 in Orlando, Florida, and revealed that traffic volumes decreased by 13.6 % amid the pandemic, leading to reduced crash frequency, while average speeds increased by 11.3 %, resulting in increased crash severities.

To address safety concerns, several studies have investigated motorcycle safety, but the majority of the studies were focused on motorcyclist crash injury severity (e.g., Savolainen and Mannering, 2007; Cheng et al., 2017; Farid and Ksaibati, 2021). For example, studies have highlighted that factors such as motorcyclists’ characteristics (age and ethnicity), crash-specific factors (collision type and speeding),

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roadway characteristics (rural roads and road surface conditions), helmet usage, and impaired driving (under the influence of drug, alcohol, or medication) significantly contributed to motorcycle crash injury severity (e.g., Savolainen and Mannering, 2007; Shaheed and Gkritza, 2014; Alnawmasi and Mannering, 2019). Motorcycle-specific safety studies, particularly focusing on motorcycle crash frequency, are rather limited, especially in the U.S. For example, Flask et al. (2014) employed Bayesian multi-level mixed effects models to predict multi-vehicle motorcycle crash frequency along road segments in Ohio. Motorcycle crash data along 32,289 state-maintained road segments in Ohio were used. The results showed that motorcycle crash frequency was associated with narrower lane and shoulder widths and increased horizontal curvature and vertical grade. Manan et al. (2013) developed safety performance functions (SPFs) for motorcycle crash casualties on Malaysian primary roads utilizing the negative binomial (NB) model. Three-year crash data between 2007 and 2009 from three different states in Malaysia were used. Their results showed that the number of access points per kilometer and the average daily number of motorcycles were significantly associated with motorcycle casualties.

Rural multilane roadways pose unique safety challenges due to their high operating speeds and limited lighting, which increases the risk of traffic crashes. Banihashemi (2015) investigated the effect of horizontal curvature on traffic crashes along rural multilane highways. Five-year crash data were used between 2007 and 2011 on rural multilane highway segments in Washington state. Their results showed that using a crash modification factor (CMF) for horizontal curvature significantly improved the crash prediction for tangent and curve segments along rural multilane highways. Asal and Said (2019) developed local SPFs for rural multilane divided highways in Egypt while applying the NB models. A total of 2,153 crashes on 17 rural multilane roadways between 2007 and 2014 were used. Their results showed that curve density (i.e., number of horizontal curves per mile), lane width, and shoulder width were significant geometric features associated with crashes on rural multilane divided highways. Overall, the review of literature has shown that there exist limited studies that specifically investigated motorcycle crashes along rural multilane roadway segments.

The NB (or Poisson-gamma) model is the most frequently used crash frequency modeling approach in transportation safety analysis (Lord and Mannering, 2010; Khaddar et al., 2023). However, the NB model is limited to over-dispersed data and cannot handle under-dispersed count data (Lord and Mannering, 2010). To address this limitation, studies have employed the Conway-Maxwell-Poisson (CMP) distribution, a flexible distribution that accounts for both over-dispersed and under-dispersed data. For example, Lord et al. (2008) developed and compared the motor-vehicle crash SPFs while applying the standard NB and CMP models. Crashes at rural four-lane highways in Texas and crashes at signalized four-legged intersections in Toronto were used. Their results suggested that CMP models performed similarly to the NB models in terms of goodness-of-fit (GOF) statistics and predictive performance. Hosseinpour et al. (2022) developed SPFs for commercial motor vehicles (CMVs) along roadway segment facilities while employing CMP-based models. The study used fatal and injury CMV crash data between 2015 and 2019 in Kentucky. The authors compared the standard NB and CMP-based models and concluded that CMP-based models performed better than the NB model.

A thorough review of the literature revealed that despite motorcycle crashes accounting for a significant portion of traffic fatalities in the U. S., there is a lack of motorcycle-specific safety studies, particularly those focusing on motorcycle crash frequency. Furthermore, the majority of motorcycle-related safety studies focused on crash injury severity, and limited studies have developed motorcycle crash-specific SPFs along rural multilane segments, especially while accounting for the pre-COVID-19 and post-COVID-19 periods separately. Please note that the term “post-COVID-19” (or post-pandemic) in this study implies “during and post COVID-19”. This study takes the initiative to address this gap by comprehensively investigating motorcycle crash frequency along

rural multilane roadway segments, considering each of the pre-pandemic (2015–2019) and post-pandemic (2020–2022) periods in Kentucky.

2. Methodology

2.1. SPF model development

The negative binomial (NB) model is commonly employed as an SPF tool; however, it is limited to handling over-dispersed count data and cannot accommodate under-dispersed count data. The CMP (or COM-Poisson) distribution generalizes the well-known Poisson distribution and accounts for both over-dispersed and under-dispersed count data (Shmueli et al., 2005). The probability mass function of a CMP distribution is shown in Equations (1) and (2).

$$P(Y = y) = \frac{\lambda^y}{(y!)^\nu Z(\lambda, \nu)} \quad (1)$$

$$Z(\lambda, \nu) = \sum_{n=0}^{\infty} \frac{\lambda^n}{(n!)^\nu} \quad (2)$$

where: Y is the discrete count, λ is a centering parameter that approximates the mean of the observations, and ν is the dispersion parameter of the CMP distribution. The condition $\nu > 1$ corresponds to under-dispersion and $\nu < 1$ to over-dispersion.

A promising generalization of the CMP model is the heterogeneous CMP (or HTCMP) model, which allows the dispersion parameter to vary across the observations as a function of site-specific characteristics. The dispersion parameter of the HTCMP model is expressed as a function of site-specific attributes, as follows:

$$\ln(\nu) = \gamma \psi_i \quad (3)$$

where: ψ_i is the vector of site-specific characteristics, and γ is the vector of regression coefficients corresponding to ψ_i . If none of the variables are significant in the dispersion part, the HTCMP model simplifies to the CMP model form, with a fixed dispersion parameter across all sites.

2.2. Goodness-of-Fit (GOF) metrics

For each of the CMP and HTCMP models in pre- and post-pandemic periods, the overall GOF measure was determined by calculating the deviance statistic (D), as shown in Equation (4).

$$D = 2^*(LL_\beta - LL_0) \cong \chi^2_{(df=p)} \quad (4)$$

where: LL_β is the log-likelihood at convergence, LL_0 is the log-likelihood of the null model (i.e., constant-only or intercept-only model), and p is the number of predictors in the final model. Other GOF measures used to assess the fitted models include the Akaike information criterion (AIC), Bayesian information criterion (BIC), and McFadden pseudo R^2 . The best-fit models have lower AIC and BIC values, while a higher McFadden pseudo R^2 suggests a superior fit. These measures are computed as shown in Equations (5) through (7).

$$AIC = -2^*LL_\beta + 2^*P \quad (5)$$

$$BIC = -2^*LL_\beta + P^*\ln(n) \quad (6)$$

$$McFaddenpseudoR^2 = 1 - (LL_\beta/LL_0) \quad (7)$$

where: P is the number of model parameters, and n is the number of observations. The likelihood ratio test (LRT) was also used to evaluate the relative performance of the fitted CMP and HTCMP models (Pathivada et al., 2024). The LRT is based on the differences in the log-likelihoods of the fitted models (LL_{CMP} , LL_{HTCMP}), which follows a Chi-squared distribution with degrees of freedom equal to the difference

between the number of estimated parameters in both models (k), see Equation (8).

$$LRT = 2*(LL_{HTCMP} - LL_{CMP}) \cong \chi^2_{(df=k)} \quad (8)$$

The predictive performance of the fitted SPFs with the validation dataset was also evaluated using the absolute deviance (MAD) and mean square prediction error (MSPE), as shown in Equations (9) and (10).

$$MAD = \frac{\sum_{i=1}^n |\mu_i - y_i|}{n} \quad (9)$$

$$MSPE = \frac{\sum_{i=1}^n (\mu_i - y_i)^2}{n} \quad (10)$$

where: y_i and μ_i are observed and predicted number of crashes at road segment i , respectively, and n is the number of sites in the validation dataset. The lower the MAD and MSPE values, the superior prediction performance is.

3. Data collection and processing

Crashes involving at least one motorcycle were classified as motorcycle-related crashes, and were identified based on the vehicle type(s) reported in the crash database. Eight years of motorcycle-involved crash data from 2015 to 2022 were collected from the Kentucky Transportation Cabinet (KYTC) crash database. The eight-year crash records were analyzed separately in two periods, i.e., pre-pandemic (2015–19) and post-pandemic (2020–2022). The roadway-specific inventory (e.g., posted speed limit, number of lanes, and shoulder width) and annual average daily traffic (AADT) were extracted from the KYTC's Highway Information Extract View and Interface (HIEVi) database (KYTC, 2024). The crash database was integrated with the HIEVi database based on the route ID and milepost of each crash. Since the present study investigates motorcycle crashes on rural multilane roadway segments, crashes on all other facilities were excluded from the analysis. After removing crash records with missing information, there were a total of 323 and 177 motorcycle crashes along rural multilane roadway segments during the pre- and post-pandemic periods, respectively.

To investigate motorcycle crash frequency along roadway segments, the road sections were split into homogeneous segments based on the homogeneity in roadway geometric and traffic characteristics. Crash frequency was then obtained by aggregating motorcycle crashes over the homogeneous segments. In this regard, the features used to produce homogeneous segments were area type (rural or urban), roadway type (undivided or divided), number of traffic lanes, posted speed limit, and AADT. A homogeneous segment ended where at least one of these

features changed, and a new homogeneous segment subsequently started.

The segmentation process was done separately for the pre- and post-pandemic crash data. A total of 277 and 173 homogeneous segments were produced for the pre- and post-pandemic periods, respectively. Note that approximately 20 percent of the total rural multilane segments had zero motorcycle-related crashes, randomly collected across Kentucky to develop valid SPFs. Fig. 1 shows motorcycle crash counts along rural multilane road segments for the pre- and post-pandemic periods. Motorcycle crash data in pre- and post-pandemic periods were under-dispersed (i.e., the crash variance is less than the mean), making CMP-based models appropriate for fitting the SPFs. Furthermore, for each period, the homogeneous segments were then randomly split into calibration (75 %) and validation (25 %) datasets, where calibration datasets were used to develop crash prediction models (i.e., SPFs) and validation datasets were used to evaluate the prediction performance of the fitted models. Table 1 shows descriptive statistics of the explanatory variables used in the motorcycle SPF development along rural multilane segments.

4. Motorcycle SPF modeling results

It should be noted that before running separate models (pre- and post-pandemic), the authors have plotted the monthly crash trend in each of the pre-pandemic (2015–2019) and post-pandemic (2020–2022) periods and different trendlines in both periods were detected, which suggested and justified the need to fit separate models. Furthermore, the LRT was used to compare the two CMP models pre- and post-pandemic, as well as the two HTCMP models pre- and post-pandemic, and significant differences in the log-likelihoods pre- and post-pandemic were observed, which again justified the need to run separate models.

The significant model estimates of fitted CMP and HTCMP models for pre- and post-pandemic periods are presented in Table 2. The deviance statistic (D) for all the fitted models was significant at the 0.1 % significance level (i.e., P -value < 0.001), indicating that the fitted models performed significantly better than their constant-only models.

The GOF measures in Table 2 indicate that the HTCMP models were superior to their CMP counterparts. Specifically, the LRT results revealed that the HTCMP model outperformed its respective CMP model for both pre- and post-pandemic periods at the 5 % significance level. Additionally, AIC and McFadden pseudo R^2 values indicate that the HTCMP models provided a better fit than their CMP counterparts. Furthermore, in terms of model validation performance, the HTCMP models exhibited the lowest MAD and MSPE values, indicating better prediction capability. The superiority of the HTCMP models implies that the varying dispersion parameter effectively captured under-dispersion at rural multilane segments. For instance, in both pre- and post-

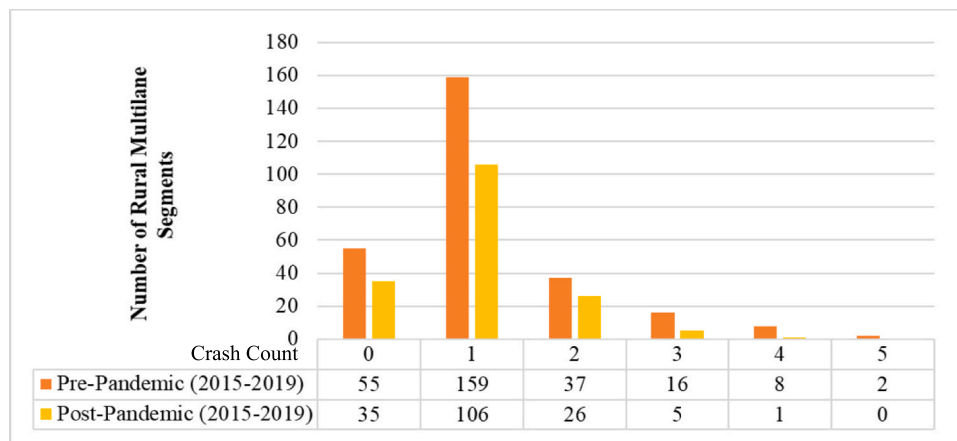


Fig. 1. Motorcycle Crash Counts along Rural Multilane Road Segments.

Table 1
Descriptive Statistics of Explanatory Variables on Rural Multilane Roadway Segments.

Variable	Variable Description	Pre-Pandemic (2015–2019)				Post-Pandemic (2020–2022)			
		No. of Sites		No. of Motorcycle Crashes		No. of Sites		No. of Motorcycle Crashes	
Calibration Dataset	Site and crash summary statistics	208		242		130		134	
Validation Dataset	Site and crash summary statistics	69		81		43		43	
Indicator Dummy Variables (Calibration dataset)									
Road Functional Class									
Interstate/Principal arterial	If true = 1, otherwise = 0	181		218		113		120	
Minor arterial	If true = 1, otherwise = 0	17		17		14		12	
Major/Minor collector	If true = 1, otherwise = 0	10		7		3		2	
Posted Speed Limit									
< 55 mph	If true = 1, otherwise = 0	24		20		10		10	
≥ 55 mph	If true = 1, otherwise = 0	184		222		120		124	
Terrain Type	If flat terrain = 1, otherwise = 0	67		72		46		48	
Horizontal Curve	If present = 1, otherwise = 0	53		71		32		36	
Vertical Grade	If present = 1, otherwise = 0	93		115		64		69	
Roadside Guardrail	If present = 1, otherwise = 0	81		90		50		54	
Undivided	If road segment is undivided = 1, otherwise = 0	28		26		20		20	
Median Barrier Type									
Cable	If true = 1, otherwise = 0	23		34		21		24	
Concrete	If true = 1, otherwise = 0	41		59		27		31	
Other types (e.g., guardrails)	If true = 1, otherwise = 0	10		14		2		2	
None	If true = 1, otherwise = 0	131		131		80		77	
Continuous Variables (Calibration dataset)									
Variable	Variable Description	Pre-Pandemic (2015–2019)				Post-Pandemic (2020–2022)			
		Min.*	Max.*	Mean	SD*	Min.*	Max.*	Mean	SD*
Ln(AADT)	Natural log. of annual average daily traffic	4.28	11.30	9.67	0.90	7.98	11.30	9.70	0.81
Lane Width (ft)	Lane width (in feet)	10	13	11.94	0.28	11	13	11.95	0.28
RSW (ft)	Right shoulder width (in feet)	0	14	9.22	2.70	0	14	9.14	2.73
Number of Lanes	Total number of lanes	3	7	4.37	0.90	3	6	4.31	0.82
Median Width (ft)	Median Width (in feet)	0	99	34.22	25.08	0	99	33.25	25.44

* Min. = minimum; Max. = maximum; and SD = standard deviation.

pandemic periods, the HTCMP models showed significant roadway features (e.g., right shoulder width, presence of cable median barrier, and flat terrain) in the dispersion part of the model.

For the pre-pandemic period, the model results showed that the presence of horizontal curves, higher AADT, and undivided roadways were positively associated with motorcycle crash frequency. In contrast, for the post-pandemic period, the model results showed that wider right shoulders, higher AADT, and undivided roadways were positively associated with motorcycle crashes along rural multilane road segments. For both pre- and post-pandemic periods, the posted speed limit (≥55 mph) was negatively associated with motorcycle crash frequency along rural multilane road segments. Additionally, the presence of median cable barriers was associated with reduced motorcycle crash frequency post-pandemic.

The model results from Table 2 were intuitive. For example, higher AADT was found to be associated with increased number of motorcycle crashes along rural multilane segments. This is expected, as AADT reflects exposure, and higher exposure results in increased motorcycle crashes. This finding is consistent with other safety studies (e.g., Manan et al., 2013; Haque et al., 2010; Ospina-Mateus et al., 2021). The posted speed limit (≥55 mph) was associated with reduced motorcycle crashes on rural multilane road segments. This might be since road segments with higher posted speed limits generally have better geometric design features (e.g., sight distance), which could contribute to the reduced motorcycle crash frequency. Similar findings were reported by Hosseinpour et al. (2022) and Hauer et al. (2004). Undivided rural multilane roadway segments were found to be associated with increased motorcycle crashes, possibly due to the absence of a physical separator median, which increases the risk of head-on collisions. This is consistent with the findings of Saeed et al. (2019).

The presence of horizontal curve was found to be associated with increased motorcycle crash frequency. This could be because the risk of losing motorcycle control increases at or nearby horizontal curves. This is in line with the findings of Flusk et al. (2014). Rural multilane

segments with cable median barriers were associated with reduced motorcycle crash frequency, likely because motorcyclists perceive the presence of median barriers as a potential hazard, prompting motorcyclists to drive cautiously; thereby, reducing the motorcycle crash frequency.

As previously indicated, post-pandemic, wider right shoulders were found to be associated with increased motorcycle crash frequency. This could be since wider shoulders may give drivers a false sense of security, encouraging them to drive faster than what the roadway conditions permit. This aligns with the findings of Shaik and Ahmed (2022), who reported that unsafe driver behavior significantly increased during and post-pandemic. To better understand the impact of right shoulder width on motorcycle crash frequency, Fig. 2 shows the plot of “normalized motorcycle crashes per segment” vs. “right shoulder width (in feet)”. For better analysis accuracy and to adjust for exposure, motorcycle crashes were normalized based on the number of segments within a specific shoulder width range. As seen in Fig. 2, there was a noticeable increase in motorcycle crash frequency along rural multilane segments with right shoulder widths 6 feet or wider.

5. High-crash location identification

The predicted crash frequencies from the best-fit HTCMP models pre- and post-pandemic periods were used to rank the high-crash rural multilane segments using the empirical Bayes (EB) method. The EB method accounts for the regression-to-the-mean (RTM) bias and estimates the expected average crash frequency by combining the observed and predicted crashes using a weight factor, which is a function of the dispersion parameter from the fitted SPF. The high-crash locations (HCLs) were then identified and ranked based on the higher-than-expected crashes. For more details on the EB method, readers are referred to the Highway Safety Manual (HSM, 2010). The identified HCLs were considerably different from the rankings based on other metrics, such as crash frequency and crash rate. This is because the EB

Table 2
Model Estimates for Motorcycle Crash SPFs along Rural Multilane Roadway Segments.

Variable	Pre-Pandemic (2015–2019)						Post-Pandemic (2020–2022)					
	CMP Model			HTCMP Model			CMP Model			HTCMP Model		
	Coef.	Std. Error	P-Value	Coef.	Std. Error	P-Value	Coef.	Std. Error	P-Value	Coef.	Std. Error	P-Value
Mean-parameter count part												
Constant	−7.200	1.221	<0.001	−1.248	0.189	<0.001	−8.450	2.242	0.001	−8.476	2.459	0.001
Posted Speed Limit (≥55 mph)	−0.873	0.408	0.032	N/S	N/S	N/S	−1.624	0.743	0.029	−1.991	0.743	0.007
Undivided	1.202	0.370	0.001	0.904	0.337	0.007	1.234	0.541	0.023	1.247	0.515	0.015
Right Shoulder Width	N/S*	N/S	N/S	N/S	N/S	N/S	0.275	0.084	0.001	0.390	0.098	<0.001
Presence of Horizontal Curve	0.673	0.226	0.003	0.839	0.235	<0.001	N/S	N/S	N/S	N/S	N/S	N/S
Ln(AADT)	0.691	0.124	<0.001	N/S	N/S	N/S	0.786	0.248	0.002	0.710	0.248	0.004
Median Barrier: Cable	N/S	N/S	N/S	N/S	N/S	N/S	−0.945	0.457	0.039	N/S	N/S	N/S
Dispersion part												
Constant	1.131	0.086	<0.001	4.041	0.683	<0.001	1.652	0.103	<0.001	0.697	0.649	0.283
Posted Speed Limit (≥55 mph)	–	–	–	0.895	0.264	0.001	–	–	–	N/S	N/S	N/S
Right Shoulder Width	–	–	–	N/S	N/S	N/S	–	–	–	0.116	0.062	0.064
Terrain Type: Flat	—	—	—	−0.221	0.101	0.030	–	–	–	−0.224	0.118	0.057
Presence of Roadside Guardrails	–	–	–	N/S	N/S	N/S	–	–	–	−0.273	0.116	0.020
Median Barrier: Cable	–	–	–	N/S	N/S	N/S	–	–	–	0.264	0.129	0.041
Ln(AADT)	–	–	–	−0.373	0.067	<0.001	–	–	–	N/S	N/S	N/S
Fixed Disp. Parameter	3.098	0.266	<0.001	–	–	–	5.220	0.538	<0.001	–	–	–
	Pre-Pandemic (2015–2019)						Post-Pandemic (2020–2022)					
	CMP Model			HTCMP Model			CMP Model			HTCMP Model		
Goodness-of-Fit (GOF) Statistics												
Number of observations	208			208			130			130		
Number of parameters	6			7			7			10		
Log-likelihood at convergence (LL_β)	−252.316			−245.120			−115.332			−109.778		
Log-likelihood at zero (LL_0)	−278.138			−278.138			−134.487			−134.487		
Deviance (D) = ($2^*(LL_\beta - LL_0)$)	51.64 (P-Value < 0.001)			66.04 (P-Value < 0.001)			38.31 (P-Value < 0.001)			49.42 (P-Value < 0.001)		
LRT (HTCMP vs CMP)	14.392 (df = 1, P-Value < 0.001)						11.108 (df = 3, P-Value = 0.011)					
McFadden pseudo R ²	0.093			0.119			0.142			0.184		
AIC (Akaike Information Criterion)	516.63			504.24			244.66			239.56		
BIC (Bayesian Information Criterion)	536.66			527.60			264.74			268.23		
MAD (Mean Absolute Deviance)	0.577			0.515			0.512			0.490		
MSPE (Mean Square Prediction Error)	0.578			0.566			0.505			0.493		

* N/S = not statistically significant at 10 % level.

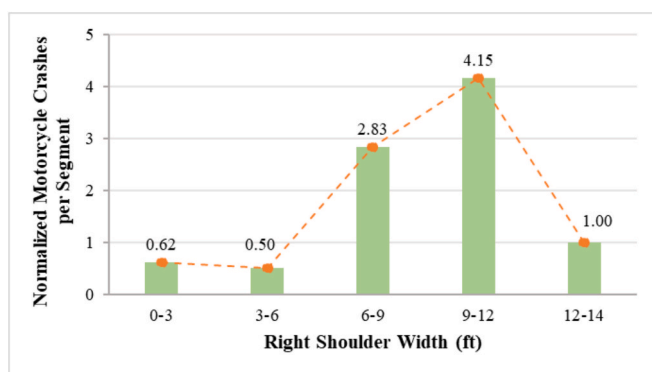


Fig. 2. Right Shoulder Width (in “feet”) versus Motorcycle Crashes per Segment.

method accounts for variations in traffic volume and other risk factors that traditional metrics overlook. As a result, the EB method provides a more precise assessment of crash risk, highlighting locations that may not appear as critical using only crash frequency and crash rate metrics.

Furthermore, an in-depth investigation of the motorcycle crashes at the top ten HCLs, pre- and post-pandemic, was conducted to identify the most frequent crash characteristics and facilitate targeted countermeasures. The analysis revealed that motorcycles were the most frequent at-fault vehicles, with “single-vehicle” and “rear-end” collisions being the most common manner of collision at the HCLs. Furthermore, examining the crash causes from the police crash narratives revealed that “lost

motorcycle control” and “failed to slow down or merge” were the most frequent crash causes. These findings highlight the susceptibility of motorcycles to crashes due to their inherent instability and emphasize the need for advanced driver assistance systems (ADAS) such as brake assist and lane change assist systems in motorcycles for improved rider safety (Anaya et al., 2017).

6. Conclusions and recommendations

This study investigated motorcycle crash frequency along rural multilane roadway segments in Kentucky by developing motorcycle-specific SPFs. Motorcycle-related crashes from 2015 to 2022 and various road-specific characteristics (e.g., posted speed limits, shoulder width, and AADT) were retrieved from KYTC and analyzed. To account for the effect of COVID-19, separate SPFs were fitted for each of the pre-pandemic (2015–2019) and post-pandemic (2020–2022) periods. CMP and HTCMP models were developed and compared while accounting for motorcycle crash under-dispersion during pre- and post-pandemic periods.

For both pre- and post-pandemic periods, the HTCMP model consistently outperformed its CMP counterpart across various goodness-of-fit measures and prediction capabilities. The superiority of the HTCMP model is likely attributed to the varying dispersion, expressed as a function of site-specific characteristics, compared to the CMP model, which has a fixed dispersion parameter. From the developed SPFs, for both pre- and post-pandemic, AADT and undivided roadways were significantly associated with increased motorcycle crash frequencies, while posted speed limits (≥ 55 mph) were associated with reduced motorcycle crash frequency along rural multilane roadway segments.

Post-pandemic, the presence of cable median barrier was associated with reduced crash frequency, whereas wider right shoulders were associated with increased crash frequency. Furthermore, the predicted crash frequencies from the fitted HTCMP models were used to identify higher-than-expected motorcycle crashes along rural multilane segments in Kentucky using the EB method.

Based on the study findings, proposed safety countermeasures to enhance motorcyclists' safety along rural multilane roadway segments in Kentucky are suggested, as follows:

- Providing right shoulder widths within 6 feet (i.e., ≤ 6 ft) along rural multilane segments since significant increase in motorcycle crashes was noticed with right shoulder widths 6 feet or wider.
- Since single-vehicle collisions were the most common crash type from the in-depth motorcycle HCL investigation, it is suggested to ensure provision of centerline grooved rumble strips along undivided rural multilane roadway stretches.
- From the in-depth motorcycle HCL investigation, it is encouraged to adopt driver assistance systems in motorcycles, including brake assist and lane change assist systems to enhance rider safety.

7. Study limitations and future scope

It is worth mentioning that the insights drawn from the motorcycle-specific SPFs along rural multilane roadway segments in this study mainly pertain to motorcycle crashes in Kentucky. Thus, caution is advised when transferring these findings to a broader context. Also, the study's focus on rural multilane roadways means that its findings might not fully apply to other roadway types, e.g., urban roads. Moreover, this paper only dealt with under-dispersed crash data (i.e., crash variance is less than its mean), which may limit the generalizability of the results to scenarios involving over-dispersed crash data (i.e., when crash variance is greater than its mean). Nevertheless, the study findings can be transferable to states/regions that exhibit similar traffic and motorcycle crash characteristics as in Kentucky. To expand upon this study, future research can compare the SPF model outcomes to rural multilane roadways in different U.S. states to assess the transferability of the proposed models.

CRedit authorship contribution statement

Bharat Kumar Pathivada: Data curation, Formal analysis, Software, Writing – original draft. **Arunabha Banerjee:** Validation, Writing – review & editing. **Kirolos Haleem:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition. **Tathagatha Khan:** Validation. **Dylan Justice:** Data curation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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